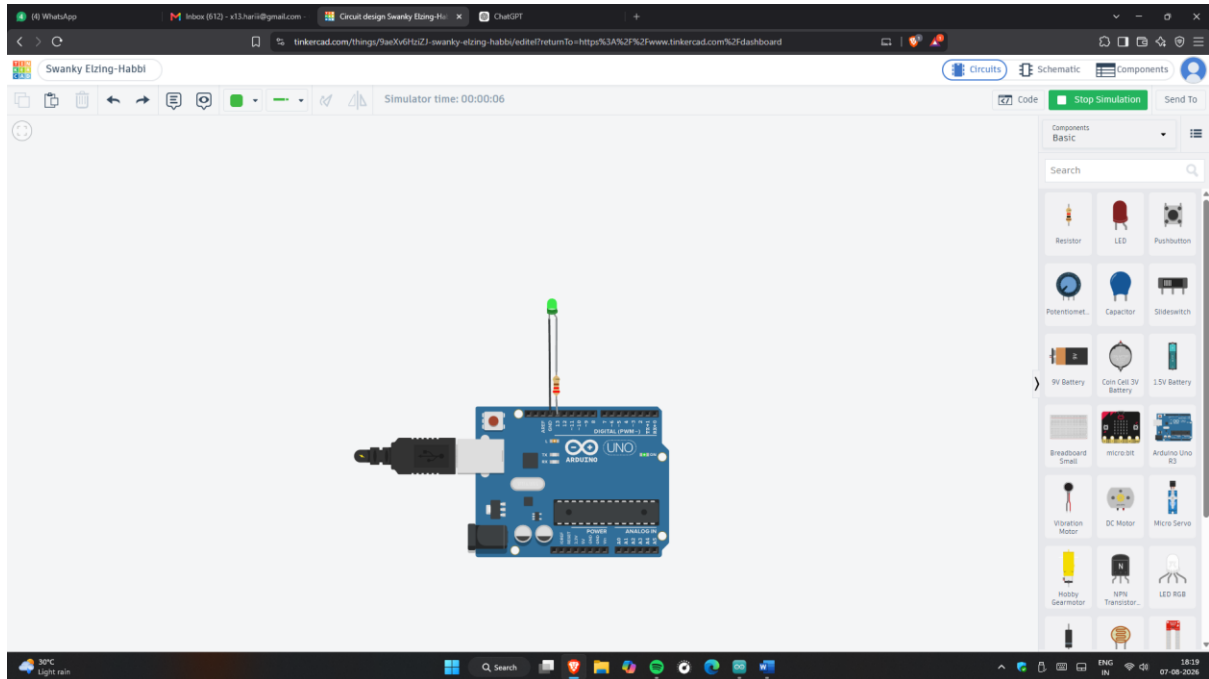


Experiment 1: LED Blinking

Arduino Experiment



CODE:

```
int ledPin = 13;

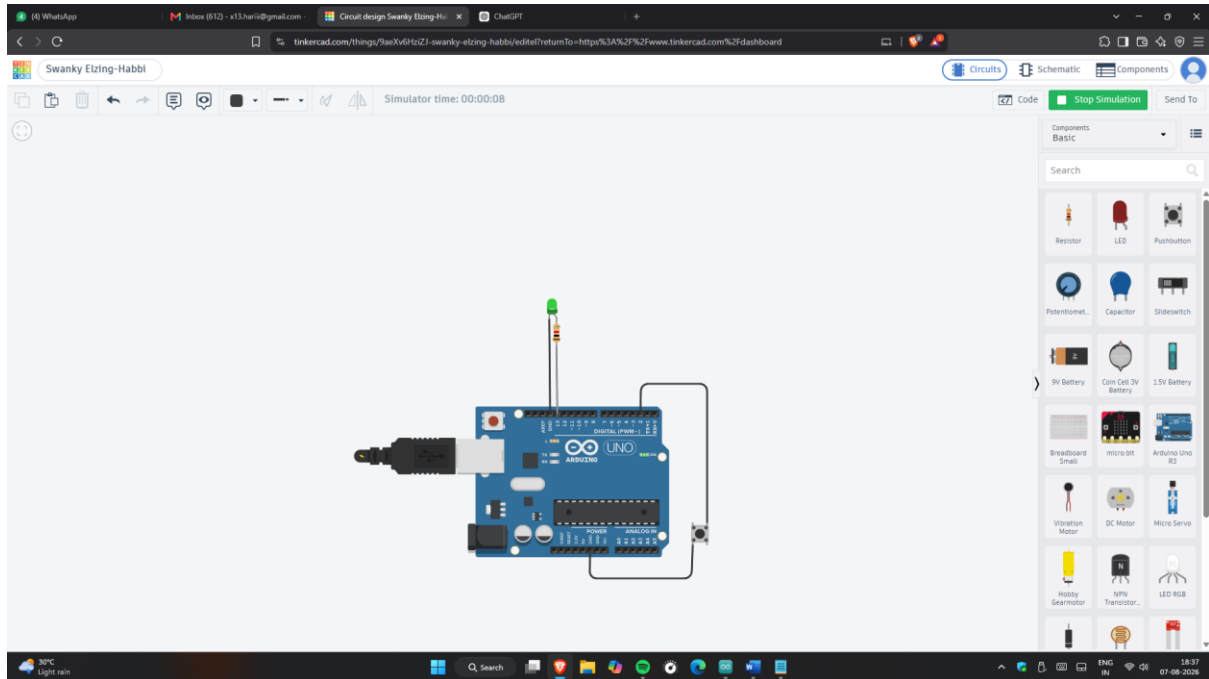
void setup()
{
  pinMode(ledPin, OUTPUT);
}

void loop()
{
  digitalWrite(ledPin, HIGH);
  delay(1000);

  digitalWrite(ledPin, LOW);
  delay(1000);
}
```

Experiment 2: Push Button with LED

Arduino Experiment



CODE:

```
int ledPin = 13;
int buttonPin = 2;

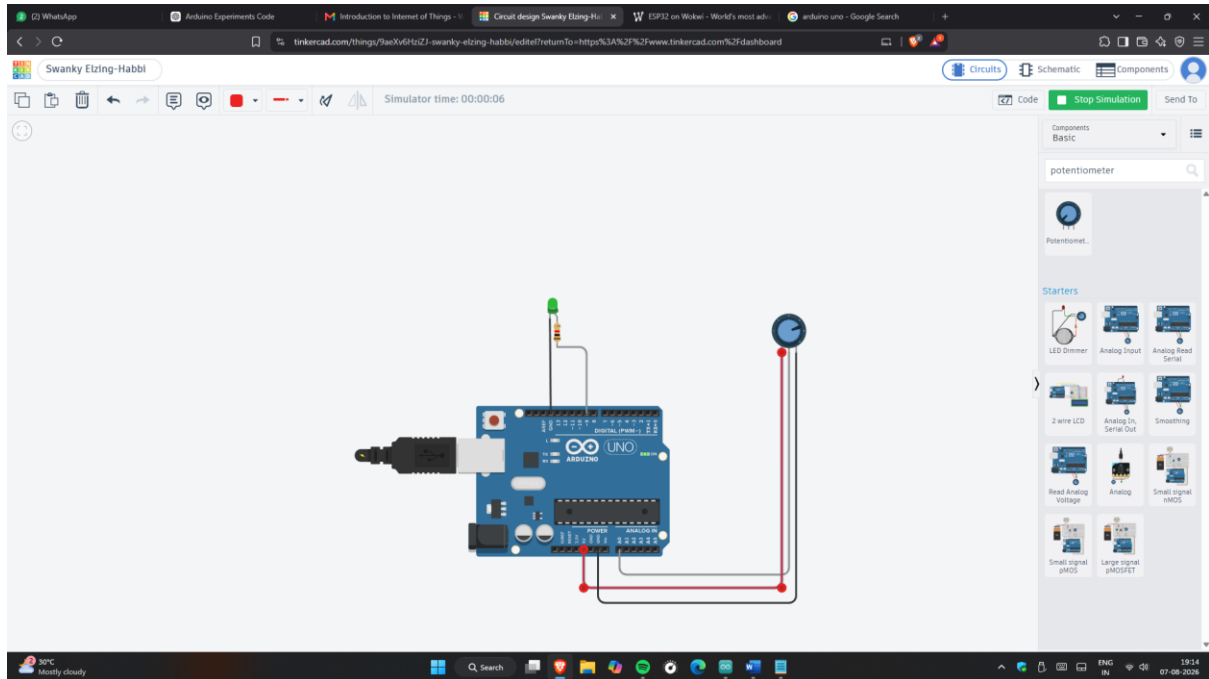
void setup()
{
  pinMode(ledPin, OUTPUT);
  pinMode(buttonPin, INPUT_PULLUP);
}

void loop()
{
  int buttonState = digitalRead(buttonPin);

  if (buttonState == LOW)
  {
    digitalWrite(ledPin, HIGH);
  }
  else
  {
    digitalWrite(ledPin, LOW);
  }
}
```

Experiment 3: Potentiometer with LED

Arduino Experiment



CODE:

```
int potPin = A0;
int ledPin = 9;

void setup()
{
  pinMode(ledPin, OUTPUT);
}

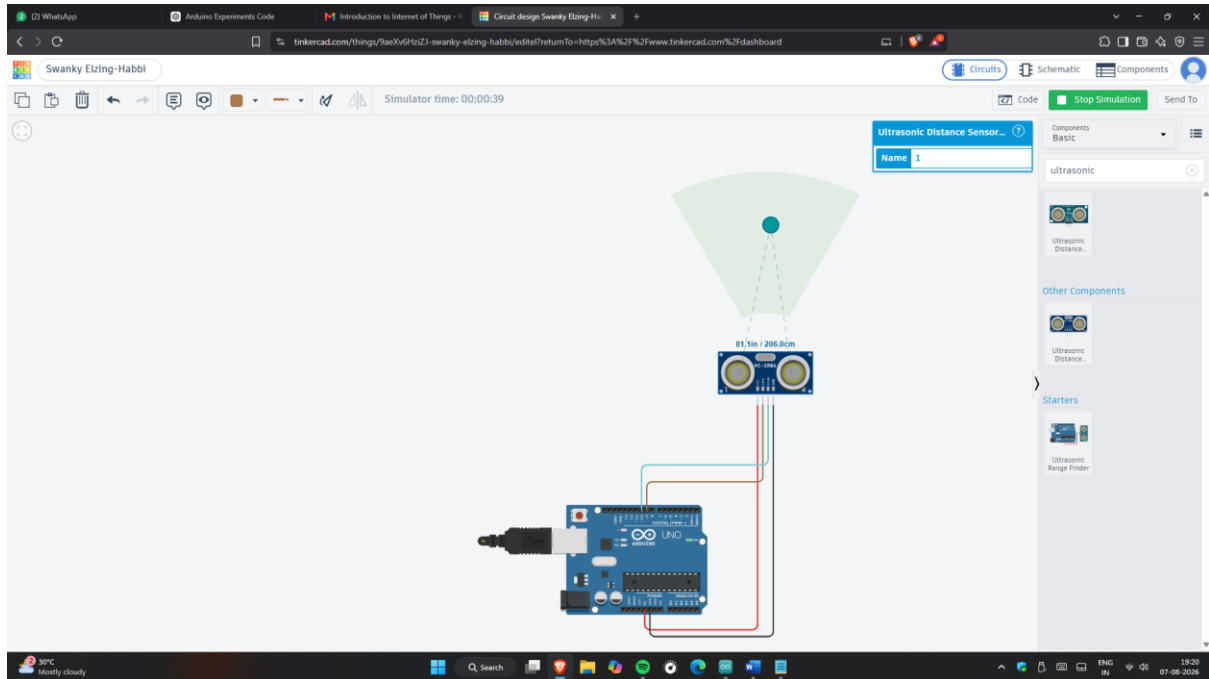
void loop()
{
  int potValue = analogRead(potPin);

  int brightness = map(potValue, 0, 1023, 0, 255);

  analogWrite(ledPin, brightness);
}
```

Experiment 4: Ultrasonic Sensor

Arduino Experiment



CODE:

```
int trigPin = 10;
int echoPin = 11;
long duration;
float distance;
void setup()
{
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

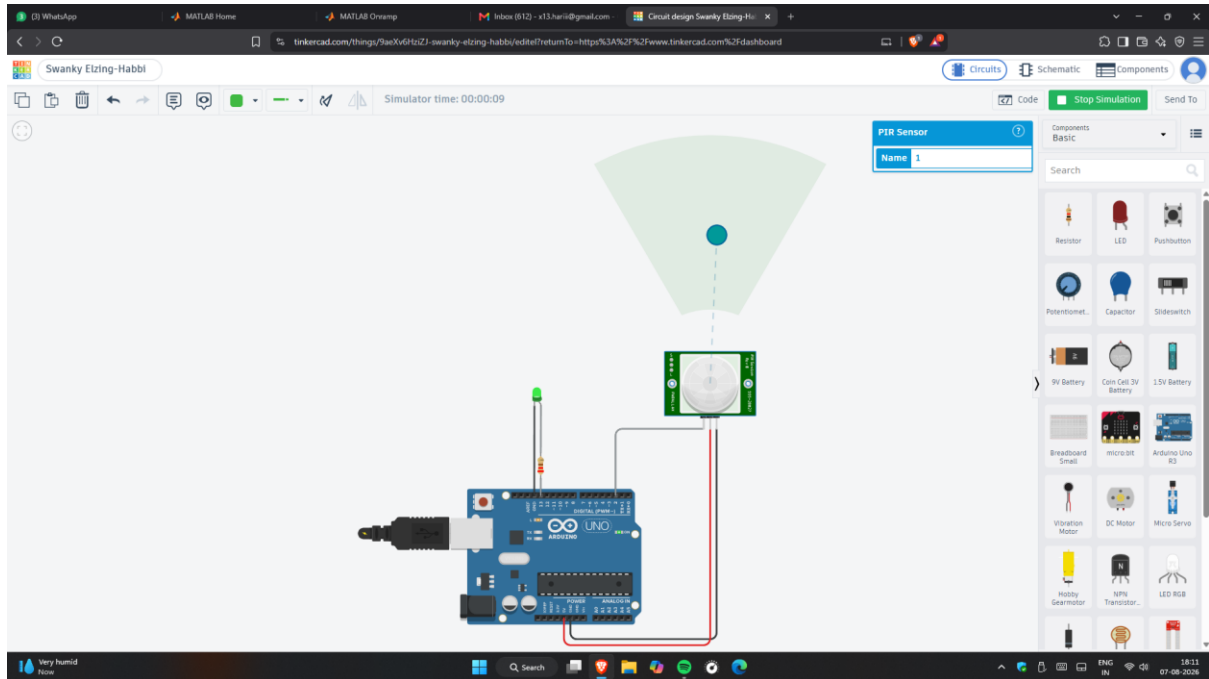
  Serial.begin(9600);
}

void loop()
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.0343 / 2;
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(500);
}
```

Experiment 5: PIR Sensor

Arduino Experiment



CODE:

```
int pirPin = 2;
int ledPin = 13;

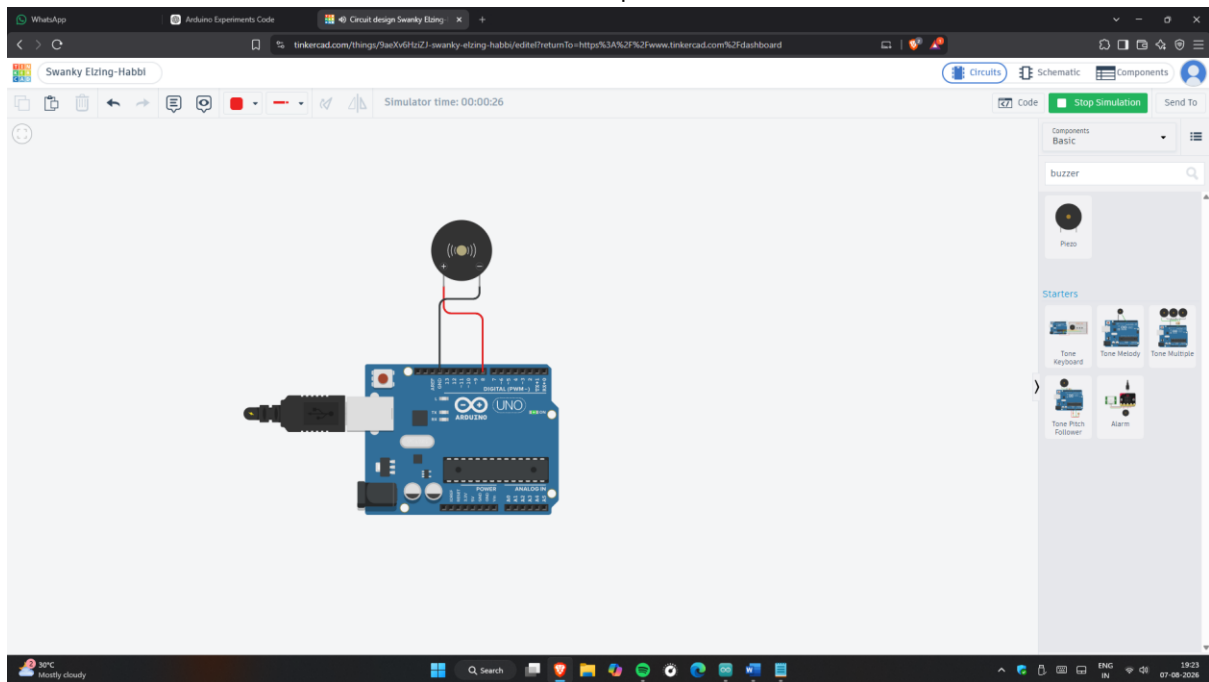
void setup()
{
  pinMode(pirPin, INPUT);
  pinMode(ledPin, OUTPUT);
}

void loop()
{
  int motion = digitalRead(pirPin);

  if (motion == HIGH)
  {
    digitalWrite(ledPin, HIGH);
  }
  else
  {
    digitalWrite(ledPin, LOW);
  }
}
```

Experiment 6: Buzzer

Arduino Experiment



CODE:

```
int buzzerPin = 8;

void setup()
{
  pinMode(buzzerPin, OUTPUT);
}

void loop()
{
  tone(buzzerPin, 1000);
  delay(500);

  noTone(buzzerPin);
  delay(500);
}
```

Experiment 7: Soil Moisture Sensor

Arduino Experiment

CODE:

```
int soilPin = A0;
int ledPin = 13;

void setup()
{
    pinMode(ledPin, OUTPUT);

    Serial.begin(9600);
}

void loop()
{
    int moisture = analogRead(soilPin);

    Serial.print("Soil Moisture: ");
    Serial.println(moisture);

    if (moisture < 500)
    {
        // Soil is dry
        digitalWrite(ledPin, HIGH);
    }
    else
    {
        // Soil is wet
        digitalWrite(ledPin, LOW);
    }

    delay(500);
}
```

Experiment 8: Temperature Sensor

Arduino Experiment

CODE:

```
int tempPin = A0;
int led = 13;

void setup() {
  Serial.begin(9600);
  pinMode(led, OUTPUT);
}

void loop() {
  int value = analogRead(tempPin);

  float voltage = value * (5.0 / 1023.0);
  float temperature = (voltage - 0.5) * 100;

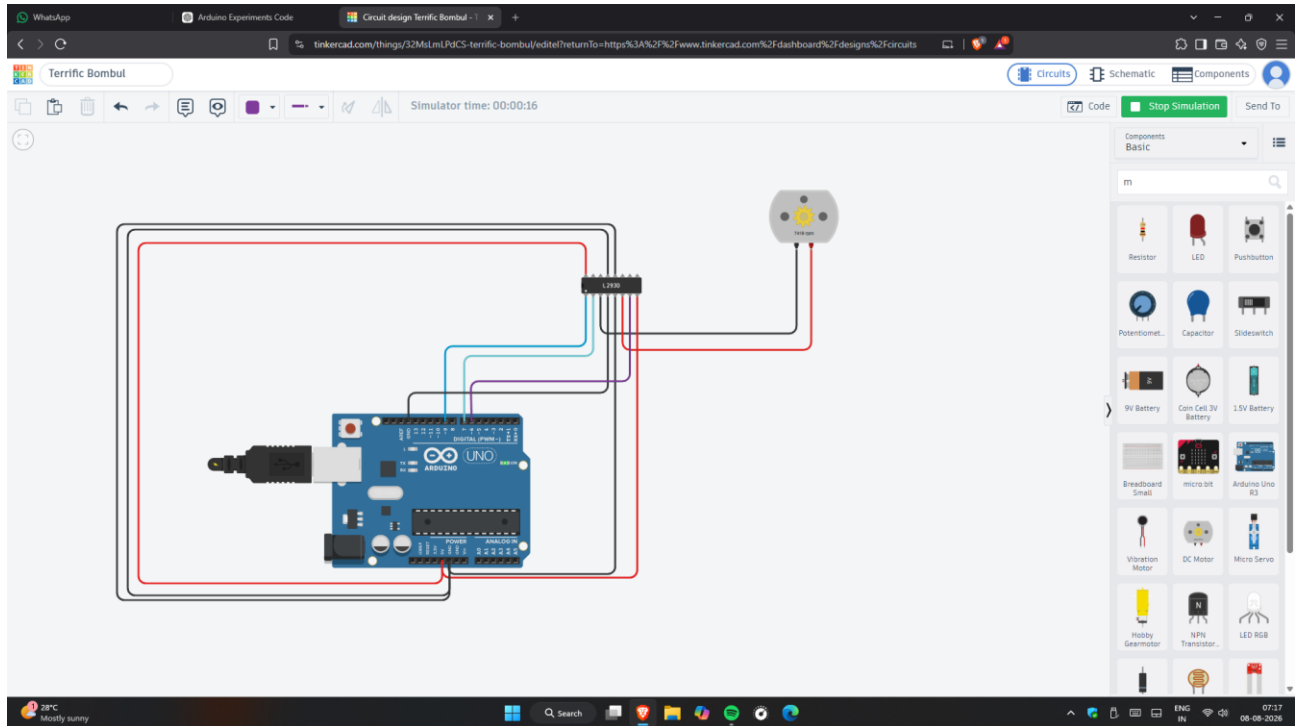
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");

  if (temperature >= 30) {
    digitalWrite(led, HIGH);
  }
  else {
    digitalWrite(led, LOW);
  }

  delay(1000);
}
```


Experiment 9: Motor

Arduino Experiment



CODE:

```
int motor = 9;

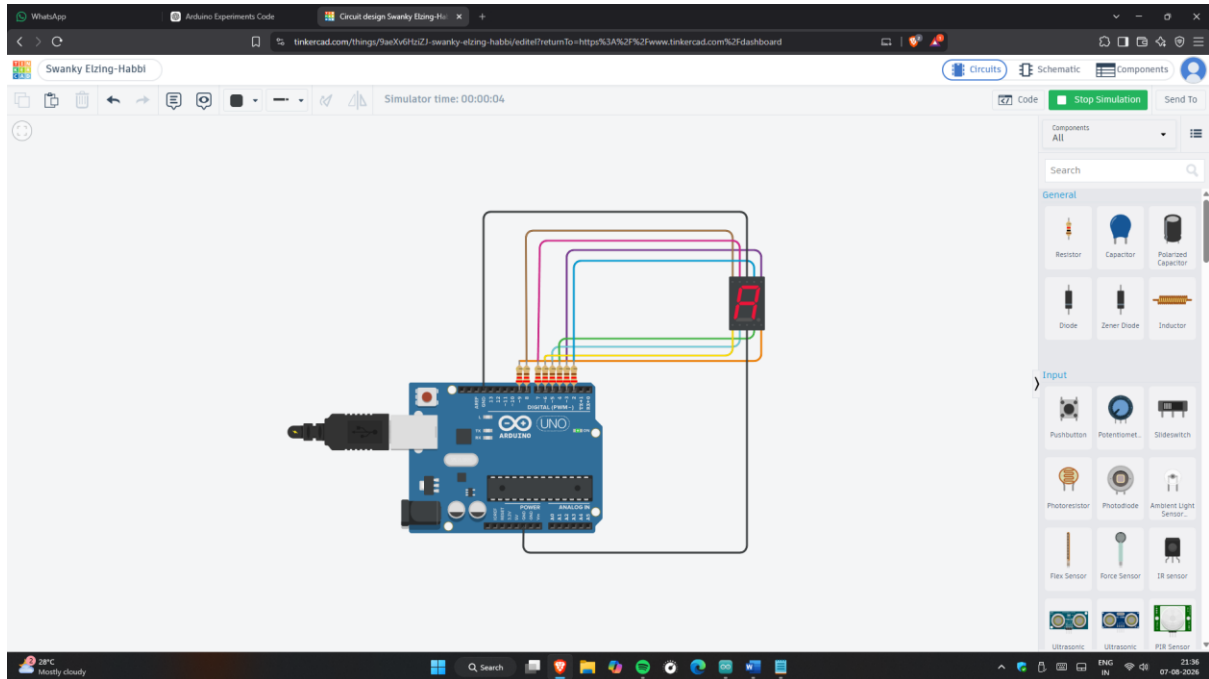
void setup() {
  pinMode(motor,OUTPUT);
}

void loop() {
  digitalWrite(motor,HIGH);
  delay(3000);

  digitalWrite(motor,LOW);
  delay(3000);
}
```

Experiment 10: Seven Segment Display

Arduino Experiment



CODE:

```
int seg[] = {2,3,4,5,6,7,8};

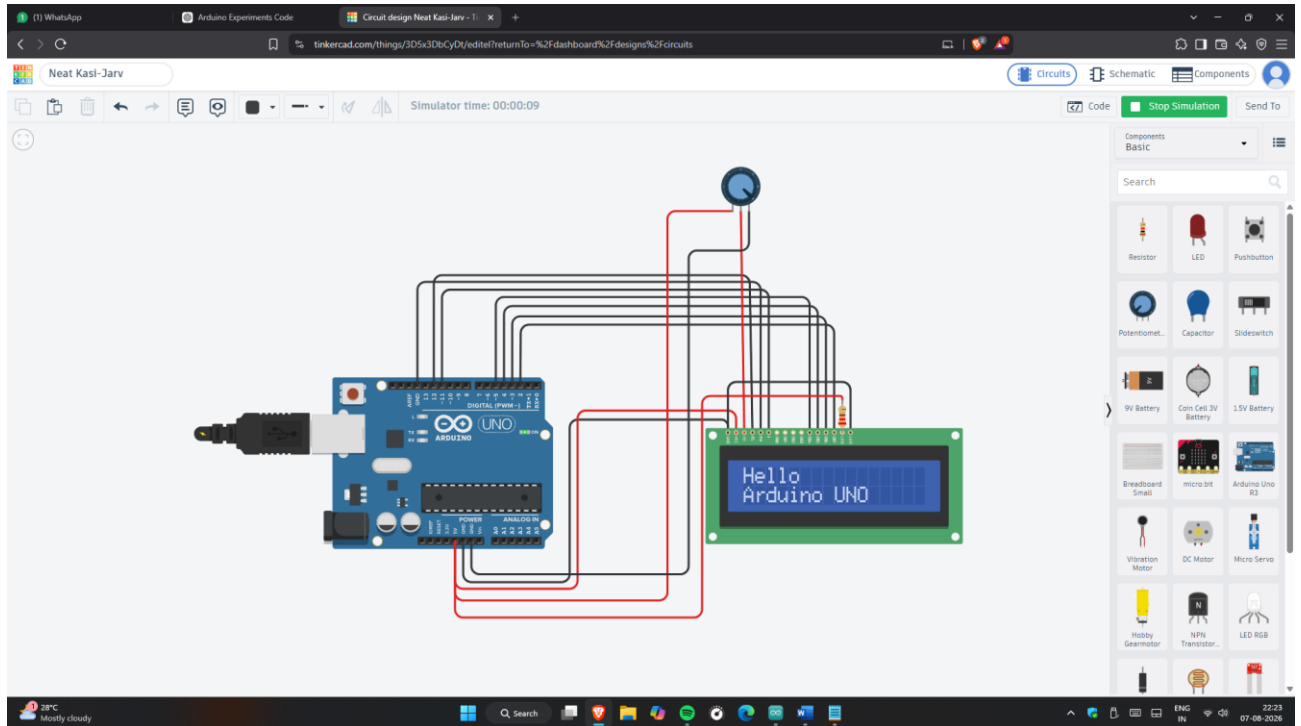
void setup() {
  for (int i = 0; i < 7; i++) {
    pinMode(seg[i], OUTPUT);
  }
}

void loop() {
  // a,b,c,d,e,f,g
  int A[] = {LOW, LOW, LOW, HIGH, LOW, LOW, LOW};

  for (int i = 0; i < 7; i++) {
    digitalWrite(seg[i], A[i]);
  }
}
```

Experiment 11: LCD

Arduino Experiment



CODE:

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(12, 11, 5, 4, 3, 2);

void setup()
{
    lcd.begin(16, 2);

    lcd.setCursor(0, 0);
    lcd.print("Hello");

    lcd.setCursor(0, 1);
    lcd.print("Arduino UNO");
}

void loop()
{
}
```